Q4:

cont... $\frac{d(\dots)}{d\mu} = \frac{-1}{2\sigma^2}\sum_{i=1}^{4} 2(x_i-\mu)\cdot -1 = 0$

$= \frac{1}{\sigma^2}\sum_{i=1}^{4}(x_i-\mu) = 0$

$= (1-\mu)+(3-\mu)+(5-\mu)+(7-\mu) = 0$

$= 16 - 4\mu \rightarrow \mu = 4$

$\frac{d(\dots)}{d\sigma^2} = \frac{-1\cdot 2\pi}{2\pi\sigma^2} - \left(\frac{-1}{2\sigma^4}\right)\sum_{i=1}^{4}(x_i-\mu)^2$

$= \frac{-2}{\sigma^2} + \frac{[(1-4)^2+(3-4)^2+(5-4)^2+(7-4)^2]}{2\sigma^4} = 0$

$= \frac{-2}{\sigma^2} + \frac{20}{2\sigma^4} = 0$

$\rightarrow \frac{-2}{\sigma^2} + \frac{10}{\sigma^4} \rightarrow \frac{-2\sigma^2}{\sigma^4} + \frac{10}{\sigma^4} = \frac{10-2\sigma^2}{\sigma^4}$

$\rightarrow 10 - 2\sigma^2 = 0$

$\rightarrow \sigma^2 = 10/2 = 5$

Q5:

$p(5|\mu) = \frac{1}{\sqrt{8\pi}} e^{\frac{-(5-\mu)^2}{8}}$

$\mu = ?$, $\sigma^2 = 4$ $\quad p(\mu) = \frac{1}{\sqrt{2\pi}} e^{\frac{-\mu^2}{2}}$

$p(\mu|x=5) = \frac{p(5|\mu)p(\mu)}{p(x=5)}$

$\propto p(5|\mu)p(\mu)$

$= \frac{1}{4\pi} e^{-\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2}}$

$\ln p(\mu|x=5) = \frac{-(5-\mu)^2}{8} - \frac{\mu^2}{2} + C$

$\frac{d(\dots)}{d\mu} = \frac{-2(5-\mu)}{8}\cdot -1 - \frac{2\mu}{2} = 0$

$= \frac{5-\mu}{4} - \mu$

$= \frac{5}{4} - \frac{5\mu}{4} \rightarrow \frac{5}{4}\mu = \frac{5}{4}$

$\rightarrow \mu = 1$

Q6:

$Var(x) = E[(X-E[X])^2]$

$= E[X^2 - 2XE[X] + (E[X])^2]$

$= E[X^2] - 2E[X]E[X] + (E[X])^2$

$= E[X^2] - 2(E[X])^2 + (E[X])^2$

$= E[X^2] - (E[X])^2$

Part 3: PCA

Q1: $Cov(X_1, X_2) = E[(X_1-\mu_1)(X_2-\mu_2)]$

$Var(u^TX) = u^T\Sigma u$

$= E[(u^TX - E[u^TX])^2]$

$= E[(u^TX - u^T\mu)^2]$

$= E[(u^T(X-\mu))^2]$

$= E[u^T(X-\mu)(X-\mu)^Tu]$

$= u^TE[(X-\mu)(X-\mu)^T]u$

$= u^T\Sigma u$

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